

District Use Case

Alamo Colleges District

Northeast Lakeview College, Universal City, Texas

Field	Value
Document status	PRELIMINARY RELEASE
Version	1.0
Date	October 7, 2026
Prepared by	Patrick Gibbs, Researcher II, National Laboratory of the Rockies
District partner	Alamo Colleges District, Facilities Operations and Construction Management
Competition	Energy District Cup, U.S. Department of Energy
Distribution	Posted to Canvas for registered EDC teams; updates will be announced through Canvas

Preliminary release: This use case is being provided so teams can begin reviewing campus conditions, assessing energy use, and identifying potential efficiency and generation opportunities. The campus utility rate structure and other items identified as pending will be provided through Canvas. Teams should document assumptions made while information is pending and revisit affected analyses when updates are issued. Financial results developed before the utility rate structure is available should be treated as preliminary.

1. Purpose and Scope of This Document

This document is the preliminary scope statement for the Northeast Lakeview College district use case in the 2026-27 Energy District Cup. It defines the district boundary, describes existing site conditions and energy performance, states the goals and constraints set by the district partner, and lists the data package that accompanies it.

Teams assigned to this use case should follow the requirements in this document the Rules Document. Submit questions through Canvas or email energydistrictcup@nlr.gov. Questions and answers will be published in a shared Q&A log so all teams receive the same information.

2. District Partner Overview

2.1 Alamo Colleges District

The Alamo Colleges District is a five-college public community college system serving San Antonio and the surrounding counties. Facilities and energy management are centralized in the district Facilities Operations and Construction Management department, which is responsible for energy management and sustainability across all campuses.

2.2 Northeast Lakeview College

Northeast Lakeview College (NLC) is the newest of the five colleges. The campus is located at 1201 Kitty Hawk Road in Universal City, Texas, at the intersection of Loop 1604 and Kitty Hawk Road. It occupies roughly 267 acres and contains approximately 370,000 square feet of academic space distributed across about ten buildings. Classes began at the current location in 2008, and the campus has continued to add buildings since, including Paluxy Hall, an approximately 90,000-square-foot STEM facility completed in 2022.

Two characteristics make the campus well suited to a district energy analysis. The buildings are relatively new and occupy a compact, contiguous area. The campus also has undeveloped acreage and large surface parking areas that teams can evaluate for generation and storage installations.

3. District Boundary and Buildings in Scope

The use case covers the NLC main campus at 1201 Kitty Hawk Road. Refer to the Google Earth file [Ref. 1] for the use case boundary and the Campus Map [Ref. 2] for building, parking lot, and infrastructure locations. Teams should evaluate energy efficiency measures for all buildings listed below and assess on-site generation opportunities within the designated boundary shown in Figure 1.



Figure 1: District Use Case Boundaries on NLC Campus [Ref. 1 NLC Google Earth.kml]

Buildings within the district boundary:

Building	Primary function	Gross floor area (sq ft)	Space Utilization	Year built	Year renovated
Salado Hall	Classrooms, laboratories, faculty offices	64,241	48.6 %	2008	N/A
Paluxy Hall	STEM laboratories, flexible learning, multipurpose	89,052		2022	N/A
Brazos Student Union	Cafeteria, Offices, Classrooms, Book Store	64,191	23.5 %	2009	2018
Blanco Hall	Fine Arts Building	25,285	35.9 %	2009	2020
Lampasas Performing Arts Center	Assembly (Auditorium, Theater)	22,320	6.1 %	2009	N/A

Building	Primary function	Gross floor area (sq ft)	Space Utilization	Year built	Year renovated
Comal Learning Resource Center	Library, Classroom, Training	82,620	35.8 %	2008	2019
Medina CTE Center	Career Technology (Classroom, Training)	22,300	20.9 %	2008	2017
JISD_JECA	Administration and Classroom Building	56,768		2008	2011
Veterans Center	Classroom, Training	13,700		2024	N/A
Llano Wellness Center	Athletic Facility, Gymnasium	47,607	24.2 %	2009	2012
Physical Plant	Utility Plant	10,763		2008	2012
Bosque Science Center	Classroom, Training, Lab Spaces	26,144	71.4 %	2008	N/A

Alamo College District has been working with consulting company Bee (building energy efficiency) to conduct a comprehensive energy audit of their buildings. The audit identified several retro Commissioning Items that could lead to cost and energy savings. The description of these items are listed in excel file NLC Retro Commissioning [Ref. 3]. Teams should quantify the cost, determine annual energy cost savings, then prioritize each of the retro commissioning items for the NLC campus.

Classrooms peak usage period is Monday through Thursday during mid-morning hours. Specialty laboratories may also be used during the evenings and weekends depending on course schedules.

Alamo Colleges conducted a space utilization study in 2022 and 2024. The percentages in the table above are from the 2022 study. Overall utilization combines the proportion of available instructional hours used with average classroom occupancy relative to the study's capacity benchmark. For example, a room used for 20 of 38 available hours per week, with an average of 25 students against a 30-student benchmark, has an overall utilization of (20 ÷ 38) × (25 ÷ 30), or approximately 43.9%.

4. District Goals and Priorities

The goals below are set by the district partner and define what a successful design looks like for this site. Teams are evaluated on how well their proposals respond to these stated priorities rather than to generic optimization targets.

4.1 Stated objectives

Alamo Colleges District's sustainability goals address economic, environmental and social well-being. Key energy related goals include:

- **Greenhouse gas emissions:** In 2009, Alamo Colleges District signed the American College & University Presidents' Climate Commitment, committing to achieve carbon neutrality by 2050. The district established an interim goal of reducing greenhouse gas emissions by 30% by 2020 and reported a 44% reduction, exceeding that target.

- **Electricity consumption**: The district aims to reduce total electricity consumption by 5% annually in accordance with Texas SB 898.

Resources:

- Alamo Colleges District Sustainability: https://www.alamo.edu/about-us/compliance/sustainability/
- Northeast Lakeview College Strategic Plan: 2025-2030

4.2 Constraints and non-negotiables

Alamo Colleges District would like proposed installations to maintain the general aesthetic of the campus. For example, at another district campus, they installed a new thermal energy storage tank instead of leaving the tank bare. They turned one side of the tank into an amphitheater where they could use the side of the tank as a backdrop or projector screen for outdoor movies and plays.

4.3 Financing posture

CPS Energy does not allow PPAs to be included in possible solar projects within its boundaries. Other than that, they do not require any extra approvals for behind the meter projects. Alamo Colleges have applied for funding for other projects through the Loan star and SECO in Texas. They applied for retro-commissioning with the following terms: 5-year equipment life and analysis period, a savings-to-investment ratio of 1.6.

5. Existing Energy Profile

This section establishes the baseline against which team designs are measured. Teams may not substitute modeled or benchmarked consumption for the actual data provided here except where explicitly noted. Teams should use the utility consumption data provided as the baseline for their analyses. Where measured data are unavailable, teams should use the organizer-provided modeled load profiles and clearly identify any additional assumptions.

Metric	Value	Source and period
Annual electricity consumption (kWh)	11,621,272 kWh	Yearly average from last 36 months
Annual peak demand (kW)	2,712 kW	Peak from last 8 months of data
Monthly peak demand profile	Included in Ref. 4	Utility Bills
Interval data availability	Included in Ref. 5	15 min data modeled based off monthly consumption
Annual natural gas consumption (therms)	3,630 CCF	Yearly average from last 36 months
Annual total energy cost	$952,475	Yearly average from last 36 months
On-site generation currently installed	There is no on-site generation currently installed.	Interview with Facilities Team

Metric	Value	Source and period
On-site storage currently installed	Thermal Energy Storage	Interview with Facilities Team

The Alamo Colleges District provided 36 months of utility data (electricity and natural gas) and 8 months of monthly electricity demand data. The competition organizers developed modeled load profiles using DOE prototype and ComStock reference profiles calibrated to the campus's monthly consumption.

NLC has a thermal storage tank on campus that they use for a utility-based demand reduction program. They have committed to reducing their peak demand through the CPS Energy Commercial Demand Response program. They are given 2-hour notice to reduce their demand, and they receive $70 per kW they reduce their demand. They are able use stored cool water to cool buildings during then specified demand reduction periods.

6. Utility Service and Rate Structure

Electric service in Universal City is provided primarily by CPS Energy, the municipally owned utility serving San Antonio and Bexar County, with Guadalupe Valley Electric Cooperative serving portions of the area east of Cibolo Creek. They have a special agreed upon rate that is a time of use rate that is not publicly posted. You can find details of the rate listed below.

Pending

Item	Detail
Electric utility	CPS Energy
Electric rate schedule	[Schedule name and effective date]
Energy charge structure	[Flat / tiered / time of use]
Demand charge structure	[$/kW, ratchet provisions, coincident vs non-coincident]
Rider and adjustment components	[Fuel adjustment, regulatory, other]
Net metering or net billing policy	[Terms for exported generation]
Interconnection process and size limits	[Utility requirements]
Available incentives or programs	[Utility rebates, demand response, efficiency programs]
Natural gas utility and rate	[Provider and schedule]

Teams should use the tariff as filed rather than a simplified blended rate. Teams should use the organizer-supplied campus rate structure for financial analysis, including applicable energy charges, demand charges, and adjustments. If a public rate source differs from the supplied campus rate, use the supplied campus rate.

7. Electrical Infrastructure and Interconnection

District scale generation and storage designs stand or fall on the electrical infrastructure they connect into. The following information is required for teams to size and site equipment credibly.

- Service entrance location, voltage, and capacity.
- Campus distribution topology: radial or looped, primary voltage, and whether the campus owns distribution beyond the meter.
- Location and rating of pad mounted transformers, switchgear, and main distribution panels.
- Available spare capacity at each building service and at the campus service entrance.
- Existing backup generation, transfer switches, and the loads they serve.
- Any known capacity, power quality, or reliability constraints reported by the utility or facilities staff.
- Single line diagrams for the campus and for individual buildings where available.

8. Authority Having Jurisdiction

8.1 Authority Having Jurisdiction (AHJ)

Northeast Lakeview College is part of the Alamo Community College District (ACCD) system, which oversees facilities planning and construction for all campuses. Teams should consult the district's Design and Construction Standards for specific requirements and guidance on design, construction, operation, renovation, modification, and general care of ACCD facilities.

The Northeast Lakeview College location is split between the jurisdictional boundaries of Universal City, Texas and the unincorporated Bexar County. Energy District Cup teams will need to take care in identifying which jurisdiction(s) their proposed project falls within, dependent on a specific building or campus area location.

For proposed projects that fall within Universal City limits, site-specific zoning and land use can be identified online in the Zoning Map. College parcels falling within the city boundary are zoned as C1 – Neighborhood Services. Land use regulations, development design standards, and building code requirements can be found within the city's Municipal Ordinance. Refer also to the city's Construction Information webpage for building permit, review process, and fee information.

Bexar County does not have zoning regulations for parcels within the unincorporated area. The Fire Marshal determines when building permits are required for construction and renovation projects. More details can be found at https://www.bexar.org/1463/Building-Permits.

The college is within CPS Energy's service area. For any proposed new energy interconnection associated with the Team's proposal, please verify requirements and review process at https://www.cpsenergy.com/content/corporate/en/construction-and-renovation/large-load-customer-growth/our-interconnection-process.html.

8.2 Airfield proximity

The campus sits near Randolph Air Force Base. Solar installations in the vicinity of active airfields can trigger glare analysis requirements and height restrictions, and the surrounding area may fall within an Air Installation Compatible Use Zone that constrains development.

9. Points of Contact and Question Process

Teams must not contact campus or district staff directly. Submit all questions about this use case through email to energydistrictcup@nlr.gov or post to Canvas. Answers will be published in a Q&A log available to all teams.

10. Data Package Contents

The following files accompany this document and are posted with it for registered teams.

Item	Format
Campus map and site plan	PDF
Building inventory with areas and construction years	XLSX
Roof plans with age and warranty status	PDF
Utility bills, 36 months, all accounts	PDF or XLSX
Interval meter data, 36 months	CSV
Applicable electric tariff	PDF
Electrical single line diagrams	PDF
Mechanical equipment inventory with age and condition	XLSX
Campus master plan	PDF
Occupancy and academic calendar schedules	PDF or XLSX